# Homework 1: DDPM Implementation

CMU 10-799: Diffusion & Flow Matching

Spring 2026

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**Due:** Saturday, January 24, 2026 at 11:59 PM ET **Total Points:** 80 + 20

**Late Due Date:** Monday, January 26, 2026 at 11:59 PM ET

**Submission:** Gradescope https://www.gradescope.com/courses/1207241

**Starter Code:** https://github.com/KellyYutongHe/cmu-10799-diffusion/

***Note:*** *By popular demand, Q6 is now an extra credit question*

## Introduction

Welcome to your first homework in CMU 10799 Diffusion & Flow Matching!

In this homework, you'll implement a Denoising Diffusion Probabilistic Model (DDPM) [Ho et al., 2020] from scratch and train it to generate realistic face images. But more than just "getting it to work," we want you to develop the intuition and debugging skills that researchers use every day. The structure of this homework mirrors a pseudo research workflow:

1. Understand your data: Before writing any model code, explore what you're trying to generate
2. Build intuition: Use the 1D playground to see diffusion in action (optional but recommended)
3. Implement: Write the core DDPM algorithm and U-Net architecture
4. Debug: Learn to diagnose common failure modes
5. Experiment: Run ablations to understand design choices
6. Reflect: Document what you learned and what you'd try next

This homework is AI-friendly. You may use any AI coding assistants, chatbots, or reference implementations. You may also use any other resources that you can find on the Internet. At the end of the homework, you'll document what resources you used.

## Part 0: Setup your codebase (0 point)

First let's get started and setup your environments and codebase by forking and cloning the startup code repository: https://github.com/KellyYutongHe/cmu-10799-diffusion.

```
git clone https://github.com/KellyYutongHe/cmu-10799-diffusion.git
cd cmu-10799-diffusion

# Run setup script (auto-detects your hardware)
./setup-uv.sh     # Recommended (faster)

# Activate environment
source .venv-*/bin/activate
```

### What You'll Implement:

| File | What to implement |
|---|---|
| `src/data/celeba.py` | Data preprocessing transforms |
| `src/methods/ddpm.py` | Full DDPM algorithm |
| `src/models/unet.py` | U-Net architecture using provided blocks |
| `configs/` | Create your own model configs |
| `train.py` | Add sampling for logging |
| `sample.py` | Add your sampling scheme |

### What We Provide:

- **U-Net** [Ronneberger et al., 2015] **building blocks** (`blocks.py`): ResBlock, AttentionBlock, SinusoidalPositionEmbedding, Downsample, Upsample, etc
- **Training infrastructure**: Mixed precision, gradient clipping, checkpointing, logging
- **EMA**: Exponential moving average for stable sampling
- **Evaluation**: KID computation via torch-fidelity [Obukhov et al., 2020]
- **Modal integration**: Cloud GPU training for those without local GPUs
- **Notebooks**: Skeleton notebook for exploration
- **Scripts**: Example scripts to run your training and evaluation jobs on a slurm cluster or on Modal

### Compute Budget:

You have **$500 in Modal credits** for the entire course (all 4 homework). Budget wisely!

## Part I: Understanding Your Data (10 points)

Since the goal for generative modeling is to model the distribution of your data, it is important to first understand what does this distribution look like.

The dataset for this course is a custom subset of CelebA [Liu et al., 2015], filtered to have specific properties. Your first task is to discover what those properties are.

### Q1. Visual Exploration [5 pts]

**(a)** [2 pts] Visualize a grid of at least 16 random samples from the training set. Include this grid below.

**(c)** [3 pts] This dataset was filtered from full CelebA using their 40 binary attributes. Based on your visual exploration, hypothesize which attributes were likely used to create this subset. What filtering criteria were used to create this subset? Why?

From the sample grid, I do not think the subset is filtered to contain only one specific demographic group. There are both male and female faces, multiple hair colors, different ages, and different expressions. However, the images are mostly clean, centered portraits with visible faces and little occlusion. My hypothesis is that the filtering removed hard cases such as hats, glasses, strong occlusions, extreme poses, or images where the face is not clearly visible. The goal seems to be a cleaner face-generation subset rather than a subset defined by a single semantic attribute.

## Q2. What did you learn? [5 pts]

**(a)** [3 pts] Imagine you found a pre-trained diffusion model that can reach SOTA performance on full CelebA (200K diverse faces). Now you generated samples and compared them to this filtered subset using KID or FID. Would you expect the FID and KID score to still be SOTA? Why?

I would not necessarily expect the FID or KID score to remain SOTA. A pre-trained model that performs well on full CelebA is trying to match the broad distribution of all CelebA faces, which includes much more variation in pose, occlusion, accessories, image quality, and facial attributes. This filtered subset appears to be a narrower distribution of cleaner, centered, less occluded face images.
FID and KID compare distributions of image features, not just whether each generated image looks realistic by itself. Therefore, even high-quality samples from a full-CelebA model may include valid CelebA faces that are outside the filtered subset distribution. In that case, the

generated distribution $p_\theta(x)$ may be close to the full CelebA distribution but still mismatched with the subset distribution, causing worse FID/KID scores against this subset.

**(b)** [2 pts] Based on your exploration, what kind of data augmentation transformation do you plan to use for your training and why?

I plan to use only mild data augmentation, mainly random horizontal flipping. The images are aligned face crops, and a horizontally flipped face should still be a realistic sample from the same face distribution. This can increase the effective diversity of the training data without changing the important facial attributes too much.
I would avoid strong rotations, vertical flips, heavy cropping, or aggressive color jitter. Those transformations could create unrealistic faces or shift the training distribution away from the filtered CelebA subset that the diffusion model is supposed to learn.

## Part II: Implement DDPM (25 points)

Now it's time to build your first diffusion model! Take a good look at the starter code and start building! Remember, you are free to add, delete and modify any and all parts of the starter code to fit your preference.

### Q3. Building Intuition [optional, 0 pts]

Before diving into the full scale training, it is generally a good idea to build some intuitions of the algorithm with some toy examples as they are easier to debug.

**(a)** [0 pts] We have prepared a Jupyter notebook *01_1d_playground.ipynb* for you to experiment with 1D toy data. This notebook contains some options of mixture of Gaussians and their visualizations, you can try out your algorithm first in this playground.

**(b)** [0 pts] Full scale training can be difficult to debug sometimes, and that's why in *train.py* we also provide an argument flag, *--overfit-single-batch*, for you to toggle an experiment where you overfit to a single batch of data for sanity checking. You should be able to iterate your model a lot faster with this experiment than full scale training.

### Q4. Implementation [25 pts]

Now implement DDPM for real! The starter code provides the skeleton, and your job is to fill in the core algorithm. Remember: this is your codebase. Feel free to modify any part of the starter code to fit your preferences. Add helper functions, reorganize files, change the config structure... whatever helps. The only requirement is that your final code runs and produces good results.

The starter code provides a detailed guideline on which files are provided and which files are the ones that need your implementation. Go check it out!

**(a)** [5 pts] Train your DDPM model to convergence. You may use either the provided configs or your own. Report:

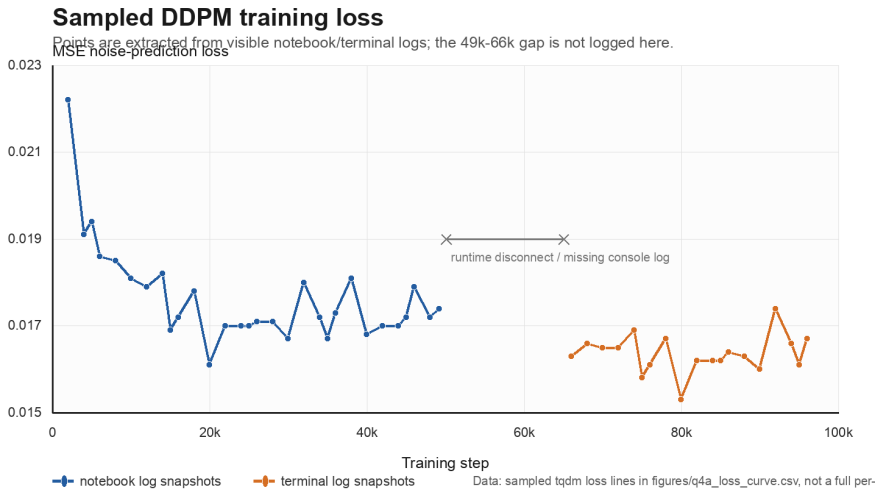
1. Model size
2. Batch size

3. Total training iterations
4. Training loss curve
5. Compute cost (GPU hours)

I trained the DDPM on the provided CelebA 64x64 subset (`electronickale/cmu-10799-celeba64-subset`) using the Colab config in `configs/ddpm_colab.yaml`. The model is a timestep-conditioned U-Net with **15,605,635** parameters (≈ 15.61M).

| | |
|---|---|
| Batch size | 64 |
| Training iterations | 100,000 |
| Image shape | $3 \times 64 \times 64$ |
| DDPM timesteps | 1000 |
| Beta schedule | linear, $10^{-4}$ to $2 \times 10^{-2}$ |
| Optimizer | AdamW, learning rate $2 \times 10^{-4}$ |
| EMA decay | 0.9999 |
| Mixed precision | enabled |
| Final checkpoint | `ddpm_final.pt` |

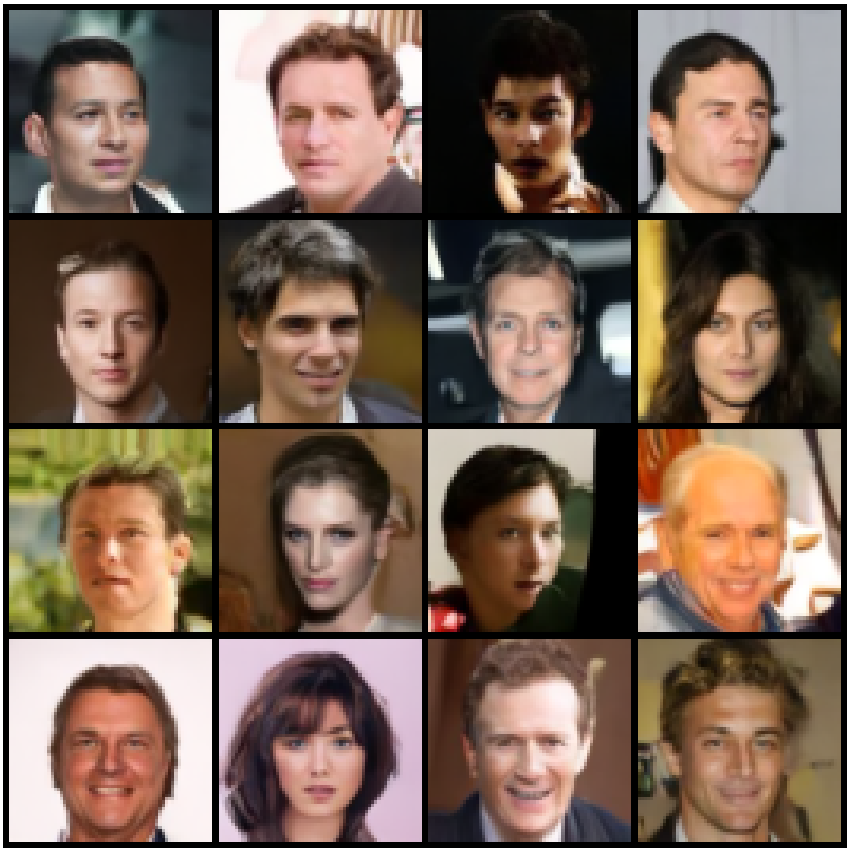
Training was interrupted by Colab runtime disconnects and resumed from saved checkpoints. The checkpoints restore model, optimizer, EMA, AMP scaler, step, and config state. The run used approximately **2.6 GPU-hours** for the actual training loop, based on the checkpoint timestamps and tqdm wall-clock output: roughly 1.5 GPU-hours for steps 0–60k and roughly 1.1 GPU-hours for steps 60k–100k. This estimate excludes package installation, dataset download, manual reconnect time, and post-training evaluation.




The plotted trace is a sampled training-loss trace rather than a full per-step metric log. The points come from the visible tqdm lines in the Colab notebook and terminal output, and the gap between the first and resumed sessions is left explicitly unconnected in the figure. The denoising MSE drops quickly in the first few thousand steps and then stays mostly around $1.6 \times 10^{-2}$ to $1.8 \times 10^{-2}$. The extracted points are saved in `figures/q4a_loss_curve.csv`.

**(b)** [10 pts] Generate a grid of 16 samples from your trained model. Include this grid below.

The following grid was generated from the final EMA checkpoint after 100,000 training iterations.



**(c)** [10 pts] Evaluate your model KID with 1k samples and report your KID score (both mean and std). You should be able to obtain KID $< 0.005$ with single L40S GPU training for a few hours.

I evaluated the final EMA checkpoint using 1000 generated samples and 1000 real images from the CelebA subset. The evaluation used `torch-fidelity` with 10 KID subsets of size 100. The generated and real image counts were both 1000.

$$\mathrm{KID}_{\mathrm{mean}} = 0.006615829467773437, \qquad \mathrm{KID}_{\mathrm{std}} = 0.001997329867895109.$$

This is slightly above the suggested 0.005 target, but it is in the same order of magnitude and the generated samples are visually face-like. The metric was computed from the checkpoint `ddpm_final.pt` with the command path `scripts/evaluate_kid.py`; the JSON output was written to `/content/kid_eval/kid_metrics.json` in the Colab runtime.

**(d)** [0 pts] Provide a zip file of your code through Gradescope "Homework 1 Code" assignment. Make sure your code is runnable because we may run it to verify your results, and do not include any large files such as model checkpoints or dataset files. If you do not provide a zip file for your code, you will receive 0 point on Part II and Part IV of this homework.

The code zip should include the runnable repository code and configs, but should **exclude** generated datasets, logs, samples, and large checkpoints such as `ddpm_final.pt`. I keep the trained model and generated figures outside the code zip and submit them only through the written report or separate artifact workflow if requested.

## Part III: Help Your "Classmate" Debug (15 points)

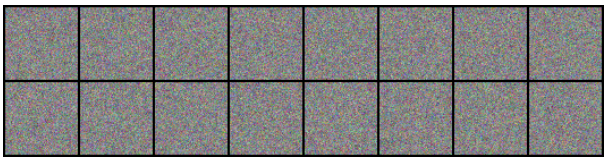
Your classmate Kale is having some trouble getting her model to work. As someone who has successfully trained a DDPM model, you decide to help!

In each of the following scenario, you may be provided a loss curve, a grid of samples or a description of the situation. Try your best to help Kale with her implementation!

### Q5. Pseudo Debugging Scenarios [15 pts]

**(a)** [5 pts] Kale trained a model for 1000 iterations and observed this seemingly nice looking loss curve. However, all her samples are pure noise. What could be the problem here? List at least two potential sources of bugs.


(a) The loss curve she observed



(b) The final samples she obtained

Figure 1: The loss curve and the samples that Kale observed.

A decreasing training loss does not guarantee that the sampling code is correct. During training, the model is only optimized to predict the Gaussian noise $\epsilon$ from $x_t$ and $t$:

$$\mathcal{L} = \mathbb{E}_{x_0,t,\epsilon}\left[\|\epsilon - \epsilon_\theta(x_t, t)\|^2\right].$$

Thus, Kale's loss curve can look reasonable even if the reverse sampling path is broken.
One likely bug is an incorrect reverse-process formula. In DDPM sampling, the predicted noise should be used to compute

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t, t)\right).$$

If the sign, coefficient, posterior variance, or timestep indexing is wrong, the sampler may fail to remove noise even though the training MSE decreases.
Another likely bug is incorrect timestep handling in the sampling loop. Sampling should start from $x_T \sim \mathcal{N}(0, I)$ and iterate backward from $T-1$ to 0. If the loop runs in the wrong

direction, has an off-by-one error, or passes the wrong timestep tensor to the model, the U-Net will denoise under a noise level different from what it saw during training.
I would also check whether the training and sampling schedules match exactly, including $\beta_t$, $\alpha_t$, $\bar{\alpha}_t$, and the number of timesteps. A mismatch means timestep $t$ has different meanings during training and sampling. Finally, Kale should verify that the sampling script loads the correct trained checkpoint and applies EMA weights if those were used for evaluation. Although 1000 iterations may be too short for high-quality face generation, completely pure-noise samples strongly suggest a sampling, schedule, timestep, or checkpoint-loading bug.

**(b)** [5 pts] Kale observed that after a while her loss would stay at a non-zero level, and it is quite "bumpy" – the loss does not decrease monotonically but instead has small fluctuations. Should she be worried about this? Why does this "bumpiness" exist? Can training for longer still be valuable for improving the model performance in her case?

She should not be worried just because the loss is non-zero and bumpy. The DDPM training objective is an expectation over the data, the timestep, and the sampled Gaussian noise:

$$\mathcal{L} = \mathbb{E}_{x_0,t,\epsilon}\left[\|\epsilon - \epsilon_\theta(x_t, t)\|^2\right].$$

Each optimization step only estimates this expectation with one minibatch, one random timestep per image, and one fresh noise tensor. Therefore, the plotted per-step loss is a noisy stochastic estimate, not the exact full objective. It is normal for it to fluctuate locally rather than decrease monotonically.
The loss also does not need to converge to zero. The model has finite capacity, the data distribution is complex, and different timesteps have different difficulty. Very noisy large-$t$ examples contain little visible image structure, while small-$t$ examples require fine local denoising. Because a batch mixes images, timesteps, and random noise, the loss can remain at a positive plateau even when the model is learning useful denoising behavior.
Training longer can still be valuable as long as the loss is not exploding or becoming NaN and the samples or evaluation metrics continue to improve. Even if the scalar MSE changes slowly, additional training may improve high-level face structure, texture details, sample diversity, EMA weights, and KID/FID. Kale should look at a moving average of the loss together with sample grids and distribution metrics instead of judging the model from raw per-iteration loss alone.

**(c)** [5 pts] After the model fully converged, Kale obtained these samples shown below. What is her bug and how to fix it?

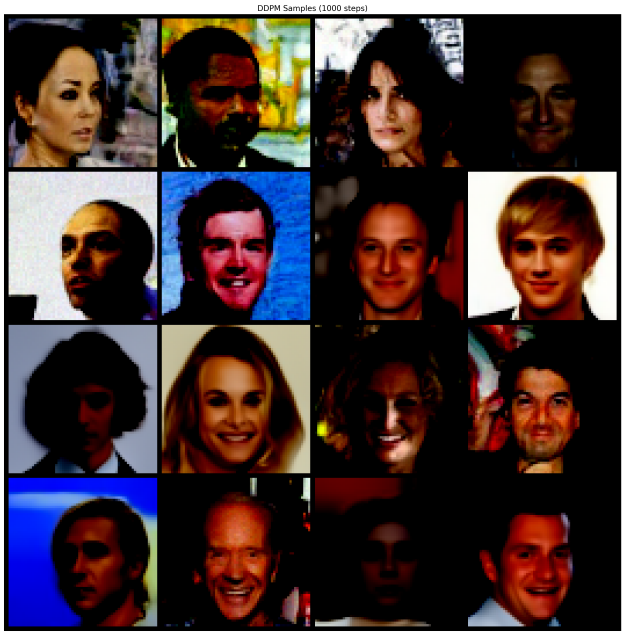



Figure 2: The samples that Kale obtained.

The samples already contain recognizable faces, so the denoising model is not completely broken. The main issue is that the saved images have an incorrect value range: large regions are clipped to black and the colors are overly contrasty. This is the typical symptom of saving tensors from the training range $[-1, 1]$ as if they were display images in $[0, 1]$.
Since the training images are normalized before entering the DDPM, the generated samples should be unnormalized before visualization:

$$x_{\text{display}} = \frac{x_{\text{sample}} + 1}{2}, \qquad x_{\text{display}} = \text{clip}(x_{\text{display}}, 0, 1).$$

If Kale skips this conversion, all negative pixel values are clipped to zero by the image-saving pipeline, which explains the dark backgrounds and saturated appearance.
The fix is to apply the inverse normalization before saving or displaying samples, for example `images = ((samples + 1) / 2).clamp(0, 1)`, or to use a helper such as `unnormalize(samples).clamp(0, 1)`. She should check the visualization code before retraining, because this bug can make a successfully trained model look much worse than it actually is.

## Part IV: Ablation Study (10 + 20 points)

In this part of the homework, we will explore different design choices of the DDPM algorithm and how they can affect the final performance.

### Q6. Alternative Parametrization [optional extra credits, 20 pts]

Note: By popular demand, this question is now an option extra credit question. I.e. You don't have to do it if you don't want to. The homework now have a total of 80 points instead of 100. If

you still complete that question you will receive 20 points extra credit.

**(a)** [7 pts] The original DDPM paper parametrized the model to predict the noise $\epsilon$. However, this is not the only option! First derive an alternative parametrization for the same model (i.e. what else can the model predict that can still yield the same mathematical formulation of the algorithm).

The forward process can be written as

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, \qquad \epsilon \sim \mathcal{N}(0, I).$$

The original DDPM parameterization trains the network to predict $\epsilon$. This is convenient because once we have $\hat{\epsilon}_\theta(x_t, t)$, we can estimate

$$\hat{x}_0 = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\hat{\epsilon}_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}},$$

and then use this estimate inside the reverse mean.
An equivalent alternative is to train the model to predict $x_0$ directly:

$$y_\theta(x_t, t) = \hat{x}_{0,\theta}(x_t, t), \qquad y = x_0.$$

This still gives the same DDPM reverse formulation because the corresponding noise prediction can be recovered by rearranging the forward-process equation:

$$\hat{\epsilon}_\theta(x_t, t) = \frac{x_t - \sqrt{\bar{\alpha}_t}\hat{x}_{0,\theta}(x_t, t)}{\sqrt{1-\bar{\alpha}_t}}.$$

Another common alternative is the velocity parameterization

$$v_t = \sqrt{\bar{\alpha}_t}\epsilon - \sqrt{1-\bar{\alpha}_t} x_0.$$

If the network predicts $\hat{v}_\theta(x_t, t)$, then both $x_0$ and $\epsilon$ can be recovered by the linear inverse:

$$\hat{x}_{0,\theta} = \sqrt{\bar{\alpha}_t} x_t - \sqrt{1-\bar{\alpha}_t}\hat{v}_\theta(x_t, t),$$

$$\hat{\epsilon}_\theta = \sqrt{1-\bar{\alpha}_t} x_t + \sqrt{\bar{\alpha}_t}\hat{v}_\theta(x_t, t).$$

Thus, velocity prediction is only a different coordinate system for the same two-dimensional linear relationship between $x_0$, $\epsilon$, and $x_t$.
A more score-matching-flavored option is to predict the score of $q(x_t|x_0)$:

$$s_t = \nabla_{x_t} \log q(x_t|x_0) = -\frac{\epsilon}{\sqrt{1-\bar{\alpha}_t}}.$$

From a score prediction $\hat{s}_\theta(x_t, t)$, we can recover

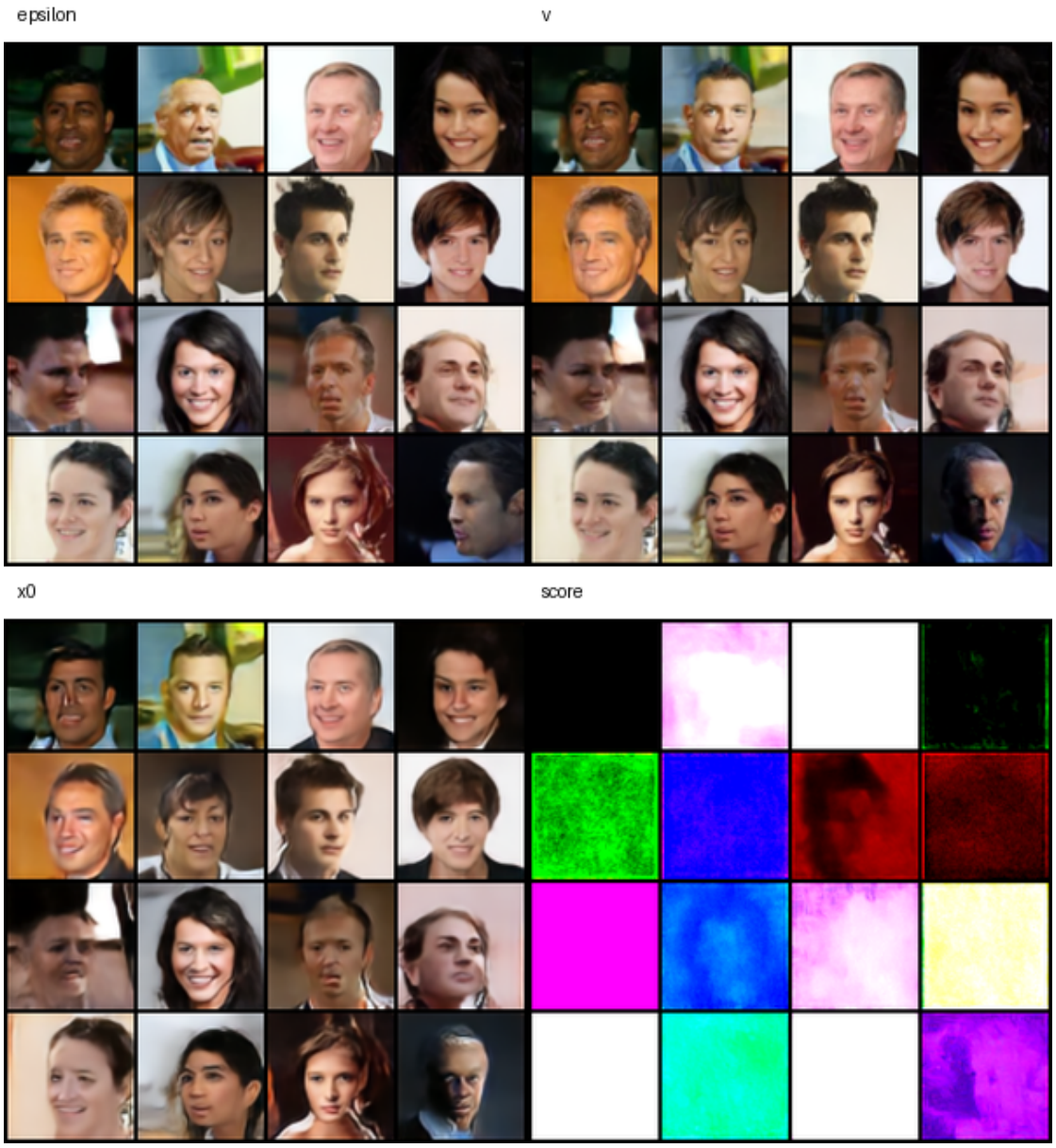
$$\hat{\epsilon}_\theta = -\sqrt{1-\bar{\alpha}_t}\hat{s}_\theta(x_t, t).$$

Therefore, $x_0$-prediction, velocity-prediction, and score-prediction can all be converted back into the noise prediction required by the DDPM reverse step. They differ in the supervised target and scaling seen during training, not in whether the reverse process can be mathematically formed.

**(b)** [10 pts] Now implement your derived parametrization and train a model with the same configuration with your previous DDPM implementation with the only difference being the model parametrization. Include a grid of 16 samples and report your KID below. (Budget tip: You can compare the two parameterizations using shorter training runs, as long as they provide meaningful comparisons.)

I implemented the alternative parameterizations in the same DDPM code path. The same timestep-conditioned U-Net architecture was trained with four prediction targets: $\epsilon$, $x_0$, $v$, and score. Across the four runs I kept the dataset, model architecture, optimizer, EMA, noise schedule, batch size, and number of iterations fixed; the only intended algorithmic change was `ddpm.prediction_type`. Each model was trained for 100,000 iterations, and KID was evaluated with 1000 generated samples against 1000 real images.

| Prediction target | KID mean | KID std |
|---|---|---|
| $\epsilon$ | 0.006382 | 0.001630 |
| $v$ | 0.008808 | 0.002058 |
| $x_0$ | 0.030816 | 0.003517 |
| score | 0.254980 | 0.009081 |

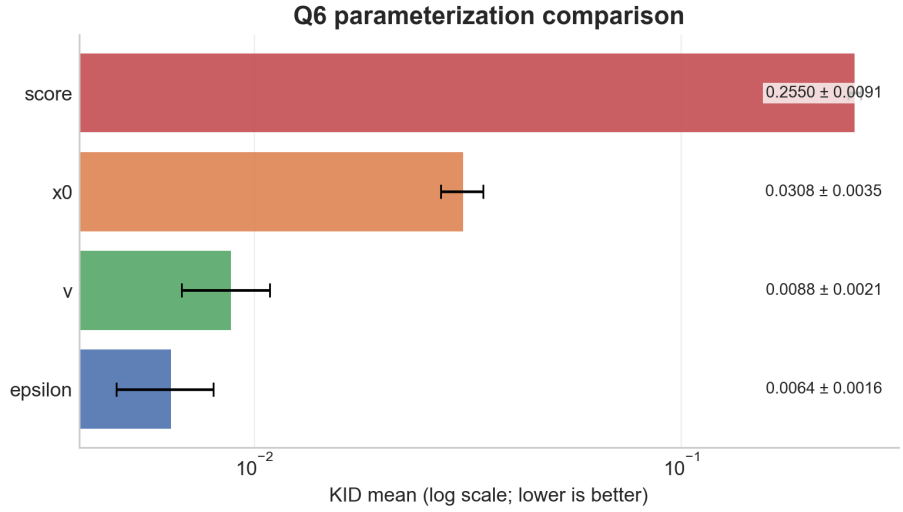


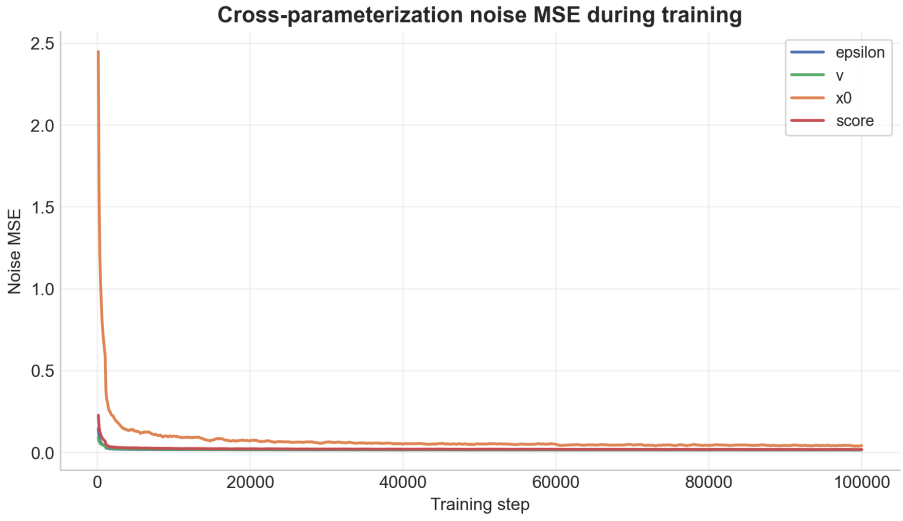

The original $\epsilon$-prediction model achieved the best KID in this run. Among the alternatives, velocity prediction was the strongest: its KID was close to the $\epsilon$ baseline and the generated samples were still face-like. Direct $x_0$-prediction was noticeably worse by KID, and the score parameterization was much worse, even though it is mathematically convertible back to a noise prediction.

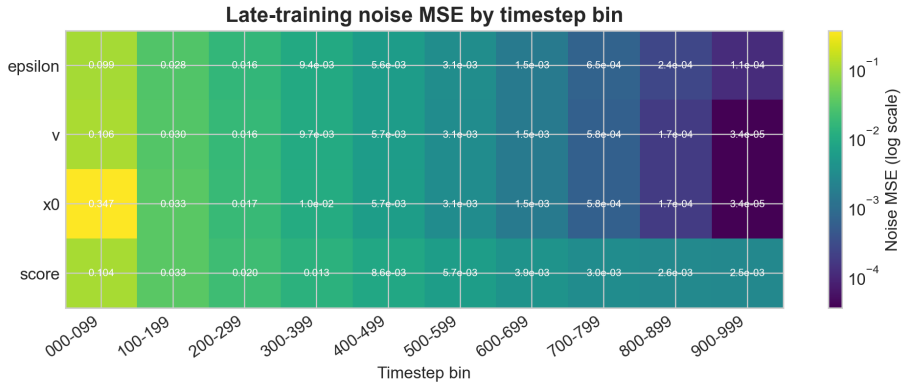
**(c)** [3 pts] Compare the performance of your new parametrization with that of the original parametrization. Share your findings below.

The main finding is that changing the prediction target changes the practical optimization problem, even when all targets can be converted back into $\hat{\epsilon}_\theta$ for the same reverse sampler. In this experiment, $\epsilon$-prediction remained the best overall choice with KID $0.006382 \pm 0.001630$. Velocity prediction was competitive with KID $0.008808 \pm 0.002058$, which supports the interpretation that $v$ is a well-conditioned coordinate system for the same $x_0$-$\epsilon$ relationship. It was not better than $\epsilon$ here, but it was close enough that I would consider it a reasonable alternative.

Direct $x_0$-prediction performed substantially worse, with KID $0.030816 \pm 0.003517$. Its final $x_0$ MSE was low because that is the direct training target, but its comparable noise MSE was worse than the $\epsilon$ and $v$ models. This matters because the reverse DDPM step still needs an accurate equivalent $\hat{\epsilon}_\theta$. The score model had the worst KID, $0.254980 \pm 0.009081$, suggesting that the score target scaling was much harder for this simple homework setup.






One useful lesson from this ablation is that raw training loss should not be compared directly across parameterizations. Each model is minimizing a different target scale. Comparable quantities such as KID, converted noise MSE, qualitative samples, and timestep-bin noise MSE give a fairer picture of which parameterization actually supports the reverse generation process.

## Q7. Sampling Steps Ablation [10 pts]

**(a)** [10 pts] DDPM typically uses 1000 timesteps for sampling, which is slow. How about using fewer steps? Report the KID scores for DDPM sampling algorithm with $100, 300, 500, 700, 900$ steps and compare with your original sampling with 1000 steps. Include 1 sample for each number of steps to qualitatively show your comparisons as well.

The sampler supports both the original full 1000-step stochastic DDPM reverse chain and a deterministic strided sampler for fewer steps. For $N < 1000$, the sampler selects $N$ descending timesteps from 999 to 0. At each selected timestep it predicts $\hat{\epsilon}_\theta$, reconstructs

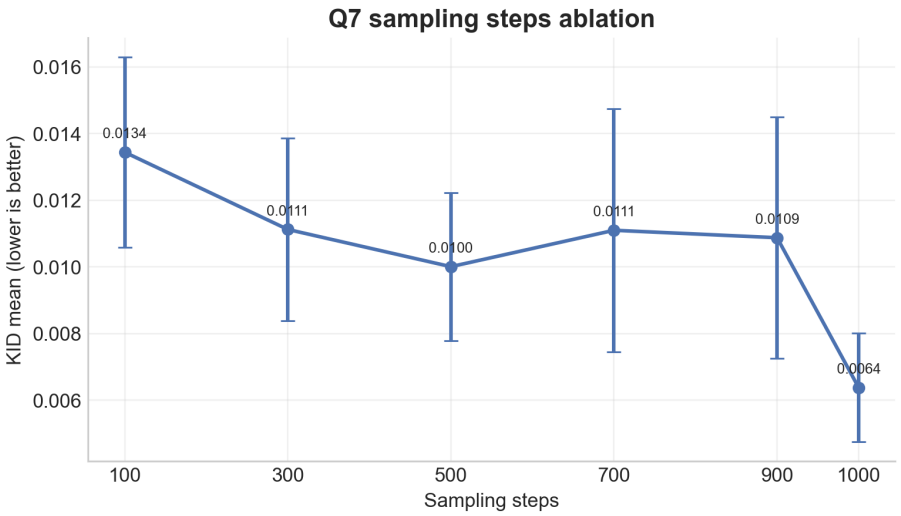
$$\hat{x}_0 = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\hat{\epsilon}_\theta}{\sqrt{\bar{\alpha}_t}},$$

and jumps to the next selected timestep $s$ with

$$x_s = \sqrt{\bar{\alpha}_s}\hat{x}_0 + \sqrt{1-\bar{\alpha}_s}\hat{\epsilon}_\theta.$$

Thus, the fewer-step sampler is a deterministic strided DDIM-style sampler using the DDPM-trained predictor, rather than the original adjacent-step posterior applied 1000 times.

| Sampling steps | KID mean | KID std |
|---|---|---|
| 100 | 0.013443 | 0.002852 |
| 300 | 0.011125 | 0.002740 |
| 500 | 0.010008 | 0.002220 |
| 700 | 0.011100 | 0.003647 |
| 900 | 0.010877 | 0.003623 |
| 1000 | 0.006382 | 0.001630 |


Q7 sampling steps ablation
KID mean (lower is better)
0.016
0.014
0.012
0.010
0.008
0.006
0.0134
0.0111
0.0100
0.0111
0.0109
0.0064
100
300
500
700
900
1000
Sampling steps


The full 1000-step sampler was still best, with KID $0.006382 \pm 0.001630$. Among the reduced-step samplers, 500 steps gave the best KID, $0.010008 \pm 0.002220$. The trend is not perfectly monotonic: 700 and 900 steps were slightly worse than 500 steps in this run. This can happen because KID is estimated from a finite sample set and because changing the stride changes the effective deterministic trajectory, not just the amount of computation.

The sample panel shows representative generated images from each sampling-step setting. I use it only as a qualitative sanity check, because the evaluation script resets the random seed for each step count. That makes the generated directories comparable and reproducible for KID, but it also means that images with the same file index across different step counts often come from the same initial noise sample. The more reliable comparison is therefore the KID table and curve above. Qualitatively, the reduced-step samples remain face-like, so the strided sampler does not completely break generation. However, the KID gap shows

the tradeoff: fewer steps make sampling faster, but the original model was trained for the 1000-step DDPM process and still benefits from using the denser reverse chain.

## Part V: Reflection (10 points)

Now that you have had a fully immersive experience with DDPM, let's take a step back and reflect on what we have learned so far.

### Q8. Reflection Questions [10 pts]

**(a)** [3 pts] Based on your experience implementing and experimenting with DDPM: What do you think DDPM is bad at? Identify 2-3 limitations you observed or suspect. These could relate to sample quality, training efficiency, sampling speed, controllability, or anything else. Be specificconnect each limitation to something concrete from your experiments or implementation.

- DDPM is expensive to tune, debug, and compare. Even in this homework-scale setup, the design space already felt large: noise schedule, prediction parameterization, sampler choice, EMA usage, and architecture details can all affect the result. In Part IV, comparing only the prediction target and the number of sampling steps already required long training runs, KID evaluation on 1k samples, and careful checkpoint handling. This makes iteration slow, and a broader model-selection process would be even more costly.
- DDPM sampling is slow. In Question 7, the best KID still came from the full 1000-step sampler, while reduced-step sampling was faster but consistently worse in quality. This shows that DDPM generation often relies on many sequential denoising steps, which makes inference expensive compared with models that generate outputs in a single forward pass.
- The optimization target is indirect and not always easy to interpret. The model does not directly learn to output a final image; instead, it learns a denoising-related target such as $\epsilon$, $x_0$, $v$, or score. The training objective is also an expectation over the data, timestep, and sampled Gaussian noise:

  $$\mathcal{L} = \mathbb{E}_{x_0,t,\epsilon}\left[\|\epsilon - \epsilon_\theta(x_t, t)\|^2\right].$$

  In practice, this means the scalar training loss is not always a transparent indicator of final generation quality. In Part IV, the raw losses across parameterizations were not directly comparable, so I had to rely on KID, qualitative samples, and converted noise MSE to judge which models were actually better.

**(b)** [3 pts] For those limitations you identified, how do you think they could be improved? Propose at least 2 concrete ideas. These don't need to be novelthey can be things you've heard about, ideas from papers, or your own speculation. For each idea, briefly explain why you think it might help. (We're not asking you to implement thesejust to think critically about the design space.)

- One improvement is to use faster samplers or distillation-based acceleration. DDIM-style deterministic sampling can reduce the number of reverse steps compared with the original DDPM sampler [Song et al., 2021], and progressive distillation is designed to compress many denoising steps into far fewer ones [Salimans and Ho, 2022]. This would directly address the slow-inference problem I observed in Question 7, where the full 1000-step sampler gave the best quality but was also the most expensive.
- A second improvement is to use better training parameterizations and variance schedules. In this homework, the choice of prediction target mattered a lot: $\epsilon$-prediction was best, $v$-prediction was competitive, and the other choices were clearly worse. This suggests that some parameterizations are easier to optimize than others. Improved DDPM design choices, such as better noise schedules and learned reverse-process variances, can make training and sampling more stable and can reduce the amount of manual trial-and-error needed [Nichol and Dhariwal, 2021].
- A third improvement is to reduce the computational burden by moving the diffusion process into a lower-dimensional latent space instead of operating on pixels directly. Latent diffusion models follow this idea and show that much of the expensive denoising work can be done in a compressed representation while still producing strong image quality [Rombach et al., 2022]. This seems especially relevant to my experience, because many of the practical difficulties in HW1 came from the cost of repeatedly training and evaluating pixel-space models.

**(c)** [3 pts] What ablations or experiments are you still curious about? List 1-2 things you would explore if you had more time and compute budget. Why do these interest you?

- I would like to run a larger parameterization study with longer training budgets and repeated seeds. In Part IV, $\epsilon$-prediction was best and $v$-prediction was close, but this conclusion came from one homework-scale training setup. I would be curious whether the ranking stays the same if the models are trained longer, if the learning rate is tuned separately for each parameterization, or if the comparison is averaged over multiple random seeds. This would help distinguish a real algorithmic advantage from a result that is partly due to limited compute or a lucky configuration.
- I would also like to study the tradeoff between sampler quality and speed more systematically. In Question 7, the 1000-step sampler was best, but 500-step sampling was the strongest reduced-step setting in this run. With more time, I would compare not only different numbers of steps but also different fast sampling methods, and I would record both KID and wall-clock generation time. This interests me because diffusion models are often limited by inference cost, so a good practical model is not just high-quality but also reasonably fast to sample from.

**(d)** [1 pts] List all the resources that you found helpful for this homework. Include AI tools, open source code, tutorials, papers, classmates that helped you, etc.

- CMU 10-799 course handout and homework specification.
- The provided starter code repository for HW1.
- The DDPM paper by Ho et al. [2020].
- The DDIM paper by Song et al. [2021].
- The improved DDPM paper by Nichol and Dhariwal [2021].
- The latent diffusion paper by Rombach et al. [2022].
- The CelebA dataset paper by Liu et al. [2015].
- The `torch-fidelity` package for KID evaluation [Obukhov et al., 2020].
- Google Colab for training and evaluation runs.
- AI tools used for explanation, debugging help, and report drafting: OpenAI Codex / ChatGPT.

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*That's it! You have completed HW 1! Congrats on your freshly baked diffusion models!*

## References

Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems*, 33:6840–6851, 2020.

Ziwei Liu, Ping Luo, Xiaogang Wang, and Xiaoou Tang. Deep learning face attributes in the wild. In *Proceedings of International Conference on Computer Vision (ICCV)*, December 2015.

Alex Nichol and Prafulla Dhariwal. Improved denoising diffusion probabilistic models. In *International Conference on Machine Learning*, pages 8162–8171. PMLR, 2021.

Anton Obukhov, Maximilian Seitzer, Po-Wei Wu, Semen Zhydenko, Jonathan Kyl, and Elvis Yu-Jing Lin. High-fidelity performance metrics for generative models in pytorch, 2020. URL https://github.com/toshas/torch-fidelity. Version: 0.3.0, DOI: 10.5281/zenodo.4957738.

Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10684–10695, 2022.

Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *International Conference on Medical image computing and computer-assisted intervention*, pages 234–241. Springer, 2015.

Tim Salimans and Jonathan Ho. Progressive distillation for fast sampling of diffusion models. *International Conference on Learning Representations*, 2022.

Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. *International Conference on Learning Representations*, 2021.